

AGNIDRISHTI AI

SIH26162 - Prototype Build Playbook

AI-Powered Geospatial Intelligence for Industrial Fire and Thermal Source Attribution

MISSION

Build a judge-convincing, end-to-end prototype between 7 and 11 September 2026 for the internal Smart India Hackathon on 12 September 2026. The prototype must demonstrate that the team understands SIH26162, can integrate real geospatial data, can classify thermal anomalies, and has a credible path to production.

Project	AgniDrishti AI
Build Team	SHARI + Omkar
Development Window	7 Sep 2026 - 11 Sep 2026
Internal Hackathon	12 Sep 2026

Core demo message: "FIRMS tells us where the heat is. AgniDrishti tells us what it most likely means, why, and whether industrial activity appears normal or anomalous."

1. Five-Day Sprint at a Glance

This is the first page both teammates should check every morning. Do not move to the next phase until the nightly milestone works end to end.

Date	Phase	SHARI	Nightly non-negotiable
7 Sep	Data + Map	React/Vite dashboard, Leaflet map, filters, markers, API wiring	Real NOAA-20/NOAA-21 FIRMS hotspot appears on your map through your FastAPI backend.
8 Sep	Geo-context	Hotspot drawer, coordinate analysis page, layer toggles, context cards	Click hotspot -> see nearby industrial features + land-cover context.
9 Sep	AI Classifier	Classification UI, confidence, "Why this classification?" panel	Click hotspot -> model returns class + probability + explanation.
10 Sep	Differentiator	History chart, persistence panel, AgniRisk UX	System distinguishes expected persistent heat from an abnormal industrial thermal event.
11 Sep	Freeze + Rehearse	UI polish, screenshots, PPT support, rehearsal	Three known demo cases work; offline cached fallback works; no major new features.

SCOPE RULE

Do not spend the five-day sprint training a giant deep-learning model. Build a real geospatial feature pipeline, a lightweight explainable classifier, temporal persistence analysis, and a reliable operations-style dashboard. That combination is far more credible for this prototype window.

2. What SIH26162 Actually Requires

AgniDrishti must not become a generic "fire markers on a map" project. The core problem is thermal-source segregation and GIS-based intelligence: distinguish industrial fire/persistent industrial heat from natural or agricultural fire using satellite thermal anomalies plus surrounding geospatial context.

- Classify / segregate industrial fires from forest or natural fires.
- Use satellite thermal detections as the trigger, not as the final answer.
- Fuse industrial infrastructure context, land cover, and historical thermal behavior.
- Store/query geospatial observations and present results as map overlays / GIS intelligence.
- Treat exact source attribution carefully: show probable nearby source and confidence, not unsupported causation.

3. Product Definition

ONE-LINE EXPLANATION

AgniDrishti takes a satellite-detected thermal hotspot and determines what the heat most likely represents - industrial fire, persistent gas flare, agricultural burning, mining activity, wildfire, or an uncertain source - using satellite thermal data, historical behavior, land cover, and nearby infrastructure.

3.1 FIRMS vs AgniDrishti

NASA FIRMS

- > Satellite detects heat
- > "There is a thermal anomaly here."

AgniDrishti

- > Hotspot detected
- > What surrounds it?
- > Is there a refinery/factory/power plant/flame/mine nearby?
- > What is the land cover?
- > Has this location been hot repeatedly?
- > Is today's FRP abnormal versus its historical baseline?
- > AI classification + confidence
- > Explanation + probable source + risk prioritization

3.2 The prototype is NOT just a latitude/longitude search

Keep coordinate analysis, but make the primary experience an operational intelligence dashboard. A judge should immediately see a map, live/recent hotspots, classifications, filters, risk, and an evidence drawer.

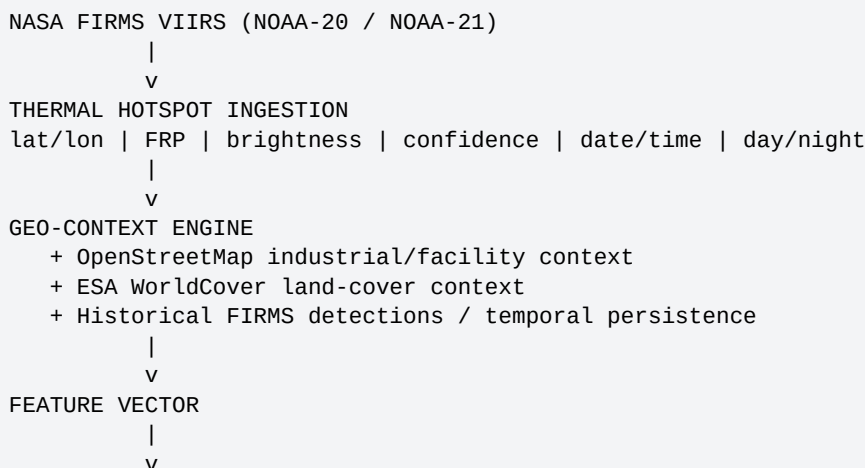
TOP CARDS: Active Hotspots | Probable Industrial | Persistent Sources | High Risk

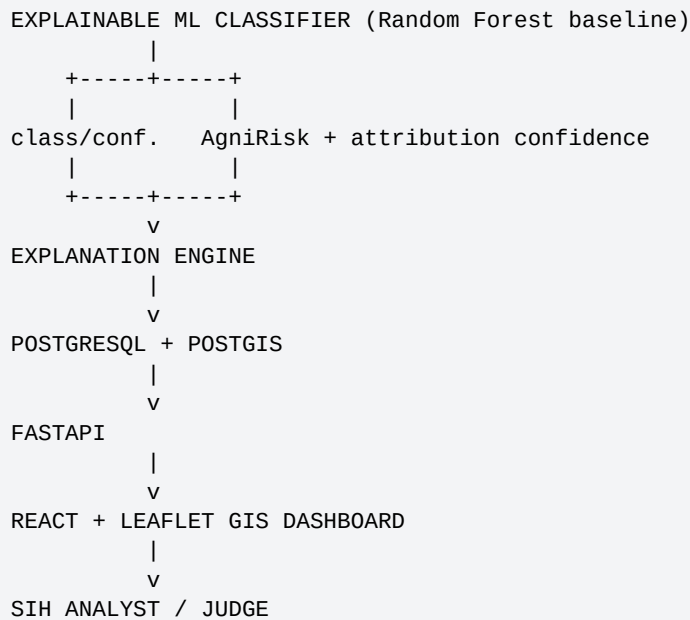
MAP: thermal markers + industrial infrastructure + land-cover/satellite overlays

FILTERS: All | Industrial | Persistent | Agriculture | Wildfire | Mining

CLICK MARKER -> classification | confidence | AgniRisk | nearby source | WHY panel

4. End-to-End Architecture





5. Data Source 1 - NASA FIRMS VIIRS

Primary detection source: use VIIRS NOAA-20 and NOAA-21 near-real-time products through the FIRMS API. The useful observation fields include latitude, longitude, I4/I5 brightness temperatures, acquisition date/time, confidence, Fire Radiative Power (FRP), day/night, and satellite.

- Obtain a FIRMS MAP_KEY before serious integration.
- Prefer NOAA-20 and NOAA-21 for the demo pipeline. FIRMS currently warns that Suomi-NPP data received after 9 March 2026 may not meet mission specifications, so do not make S-NPP your primary source for this prototype.
- Never interpret a VIIRS point as the exact burning object. NASA notes that a VIIRS active-fire detection is associated with an approximately 375 m pixel; the fire may occupy only part of that pixel. Source attribution is therefore probabilistic/contextual.

JUDGE-SAFE WORDING

Say "nearest probable industrial source" and show attribution confidence. Do not say "this factory caused the fire" unless you have independent ground truth.

6. Data Source 2 - OpenStreetMap / Overpass

For each hotspot, create a local search buffer (prototype: roughly 1-2 km, adjustable by context) and query relevant OSM features. Compute the nearest distance to each important feature type.

```

Useful OSM context tags
- landuse=industrial
- man_made=works
- man_made=flare
- power=plant
- landuse=quarry
- industrial=*
- product=*

```

- works=*

- Nearest factory / works
- Nearest industrial land-use polygon
- Nearest gas flare
- Nearest power plant
- Nearest quarry / mine
- Distance to non-industrial context where useful

Example context result
 Hotspot: 19.04131, 72.88210
 Nearest refinery/works: 380 m
 Industrial land-use: 110 m
 Gas flare: 620 m
 Power plant: 4.2 km
 Forest: 12.4 km
 => Strong industrial context

7. Data Source 3 - ESA WorldCover

Use ESA WorldCover as the land-cover evidence layer. It provides global 10 m land-cover classes and helps distinguish cropland, tree cover, built-up areas, bare/sparse vegetation, water and other surface types.

Tree cover + no nearby industry + transient hotspot -> probable wildfire
 Built-up/industrial context + nearby facility + abnormal FRP -> probable industrial event
 Cropland + no industrial infrastructure -> probable agricultural burning

- For a five-day prototype, avoid depending on a complicated Google Earth Engine authentication path if it slows you down.
- Use a practical accessible path such as a public STAC/catalog or map service for the prototype. Public Terrascope WMS/WMTS layers can be used without authentication where available.

8. Key Differentiator - Temporal Persistence

This is the feature that should make AgniDrishti more than a standard FIRMS dashboard. Ask whether the same location is thermally active repeatedly and whether today's FRP is normal for that location.

CASE A - EXPECTED PERSISTENT SOURCE
 Repeated detections at same location across many dates
 + relatively stable FRP
 + refinery/flare context
 => Persistent Industrial Thermal Source / Gas Flare

CASE B - ABNORMAL INDUSTRIAL EVENT
 Normally quiet location
 + sudden strong hotspot
 + current FRP far above historical median
 + industrial context
 => Probable Abnormal Industrial Thermal Event / Industrial Fire

CORE USP

AgniDrishti combines spatial context + temporal persistence + thermal intensity to distinguish expected persistent industrial heat from potentially abnormal industrial thermal events.

9. AI / ML Model

Use a RandomForestClassifier as the prototype baseline. It trains quickly, accepts mixed engineered features, returns probabilities, and is easy to explain. The point is not model glamour; the point is a credible geospatial intelligence pipeline.

9.1 Proposed feature vector

- bright_ti4
- bright_ti5
- frp
- confidence
- day_night
- nearest_industry_distance
- nearest_factory_distance
- nearest_flare_distance
- nearest_powerplant_distance
- nearest_quarry_distance
- forest_distance / natural-context distance
- cropland distance or land-cover class
- landcover_class
- detections_last_7_days
- detections_last_30_days
- unique_detection_days_30
- historical_median_frp
- current_frp_vs_baseline
- persistence_score

9.2 Output classes

- 0 -> Industrial Fire / Abnormal Industrial Thermal Event
- 1 -> Persistent Industrial Thermal Source / Gas Flare
- 2 -> Agricultural Burning
- 3 -> Wildfire / Natural Fire
- 4 -> Mining / Quarry Thermal Activity
- 5 -> Other / Uncertain

10. Training Labels - Use Weak Supervision, Do Not Fake Ground Truth

A technically strong judge may ask where the labels came from. Be explicit: the prototype can use geospatial weak supervision / pseudo-labeling to create an initial "silver" dataset. Do not claim thousands of manually validated industrial-fire labels if you do not have them.

```
OSM flare nearby + repeated detections -> Persistent Gas Flare
WorldCover cropland + no industry -> Agricultural Burning
WorldCover tree cover + no industry -> Wildfire
Quarry nearby -> Mining / Quarry Activity
Industrial context + low persistence + abnormal FRP -> Probable Industrial Fire
```

WHAT TO SAY

"For the prototype, we use geospatial weak supervision to construct the initial training dataset. A production version would incorporate validated incident databases and analyst-confirmed labels."

11. AgniRisk - Prototype Prioritization Score

AgniRisk is not an official emergency severity rating. It is a prototype prioritization score to help an analyst sort hotspots.

AgniRisk combines:

- Industrial proximity
- FRP anomaly
- Thermal intensity
- Temporal behavior / persistence
- Land-cover consistency
- Model probability

Example interpretation

23/100 LOW -> Expected persistent thermal source

67/100 MODERATE -> Abnormal industrial thermal activity

91/100 CRITICAL -> Probable transient industrial fire

12. Explainability Panel - Must Build

A judge should be able to click a hotspot and immediately understand why the system produced the classification.

AGNIDRISHTI ANALYSIS

PROBABLE INDUSTRIAL FIRE

AI Confidence: 87%

AgniRisk: 91 / 100

WHY THIS CLASSIFICATION?

- [+] Industrial zone detected 280 m away
- [+] Petrochemical facility detected 410 m away
- [+] Current FRP 2.4x historical median
- [+] Low persistence at this location
- [+] Land cover: Built-up
- [+] No strong forest context nearby

Nearest probable source: Example Refinery - 410 m

Attribution confidence: MODERATE

Satellite: NOAA-21 VIIRS | FRP: 48.3 MW | Brightness: 367 K

13. Exact Prototype Screens

13.1 Screen 1 - Intelligence Dashboard

- Top cards: Active Hotspots, Probable Industrial, Persistent Sources, High-Risk Events.
- Large interactive GIS map.
- Layer toggles: FIRMS Hotspots, Industrial Infrastructure, ESA Land Cover, Satellite/Base Imagery.
- Classification filters: All, Industrial, Persistent, Agriculture, Wildfire, Mining.

- Marker click opens analysis drawer with AI class, confidence, AgniRisk, source context and reasons.

13.2 Screen 2 - Analyze Location

```
ANALYZE THERMAL LOCATION
Latitude: [ 19.0412 ]
Longitude: [ 72.8791 ]
Search radius: [ 2 km ]
[ ANALYZE WITH AGNIDRISHTI ]

Processing steps
[+] Retrieving FIRMS observations
[+] Searching industrial infrastructure
[+] Retrieving land-cover context
[+] Analyzing temporal history
[+] Building feature vector
[+] Running classifier
[+] Computing AgniRisk
-> Analysis complete
```

14. Map / GIS Layers

- Base 1: OpenStreetMap.
- Base 2: Satellite / Sentinel-derived imagery where practical.
- Overlay 1: NASA FIRMS thermal hotspots.
- Overlay 2: OSM industrial infrastructure / facilities.
- Overlay 3: ESA WorldCover land-cover context.

This immediately communicates the expected multi-source geospatial fusion: FIRMS + OSM + satellite/land-cover context.

15. GIS Database - PostgreSQL + PostGIS

Use PostGIS so you can truthfully present AgniDrishti as a geospatial intelligence platform rather than a front-end map that stores marker JSON.

```
hotspots
- id
- latitude / longitude / geom
- acquisition_time
- satellite
- frp
- bright_ti4 / bright_ti5
- confidence
- daynight

facilities
- id / osm_id / name / facility_type / geom / tags

hotspot_features
- hotspot_id
- industry_distance / flare_distance / powerplant_distance / quarry_distance
- landcover
- persistence_7d / persistence_30d
- historical_frp / frp_anomaly

classifications
- hotspot_id
```

- classification
- model_probability
- agni_risk
- attribution_confidence
- model_version
- explanation_json

16. Technology Stack

Frontend	React, Vite, Tailwind CSS, React-Leaflet, Leaflet, Recharts, Axios
Backend	Python, FastAPI, Pandas, GeoPandas, Shapely, Requests, Scikit-learn, Joblib
GIS/Data	OpenStreetMap, Overpass API, ESA WorldCover, NASA FIRMS, PostgreSQL/PostGIS
ML	Random Forest, feature engineering, weak supervision, temporal anomaly analysis
Deployment	Frontend: Vercel; Backend: Render/equivalent; DB: PostgreSQL/PostGIS
Resilience	Cached real observations + DEMO_MODE fallback

17. Repository Structure

```

agnidrishti-ai/
|-- frontend/
|   |-- src/
|   |   |-- components/
|   |   |   |-- Map.jsx
|   |   |   |-- HotspotMarker.jsx
|   |   |   |-- AnalysisDrawer.jsx
|   |   |   |-- StatsCards.jsx
|   |   |   |-- LayerControl.jsx
|   |   |   |-- RiskBadge.jsx
|   |   |-- pages/
|   |   |   |-- Dashboard.jsx
|   |   |   |-- Analyze.jsx
|   |   |-- services/api.js
|   |   |-- App.jsx
|   |-- package.json
|
|-- backend/
|   |-- app/
|   |   |-- main.py
|   |   |-- api/ (hotspots.py, analysis.py, stats.py)
|   |   |-- services/
|   |   |   |-- firms.py
|   |   |   |-- osm.py
|   |   |   |-- worldcover.py
|   |   |   |-- persistence.py
|   |   |   |-- classifier.py
|   |   |   |-- risk_engine.py
|   |-- models/

```

```

| | `-- database/
| `-- requirements.txt
|
|-- ml/
| |-- notebooks/
| |-- data/
| |-- features.py
| |-- weak_label.py
| |-- train.py
| |-- evaluate.py
| `-- agnidrishti_rf.joblib
|
|-- docs/ (architecture.png, pipeline.png, demo_cases.md)
|-- sample-data/
|-- .env.example
|-- .gitignore
`-- README.md

```

18. Backend API Contract

18.1 GET /api/hotspots

Parameters: bbox, days, sensor

Purpose: fetch/return normalized thermal detections for the visible map area.

18.2 POST /api/analyze

Request

```

{
  "latitude": 19.041,
  "longitude": 72.879
}

```

Response (shape)

```

{
  "classification": "Probable Industrial Fire",
  "confidence": 0.87,
  "risk_score": 91,
  "nearest_industry": {"name": "Example Refinery", "distance_m": 410},
  "landcover": "Built-up",
  "frp": 48.3,
  "persistence_30d": 0.07,
  "reasons": [
    "Industrial facility within 500 m",
    "FRP substantially above historical baseline",
    "Low thermal persistence",
    "Built-up land cover"
  ]
}

```

18.3 GET /api/hotspots/{id}/history

Return the temporal detections / baseline metrics required for the history chart and persistence analysis.

18.4 GET /api/stats

Return the dashboard summary counts. Keep the API set small and reliable.

19. Detailed Five-Day Build Plan

19.1 7 September - Phase 1: Data + Map

SHARI

- Create React/Vite frontend skeleton.
- Build navbar, stat cards, Leaflet map, sidebar, filters, and temporary markers.
- Create a simple API service layer so data source can switch from dummy to backend cleanly.

OMKAR

- Create FastAPI backend.
- Configure FIRMS MAP_KEY safely in environment variables.
- Implement NOAA-20 and NOAA-21 FIRMS ingestion.
- Merge and normalize fields into a clean JSON contract.
- Expose GET /api/hotspots.

Night integration

React -> FastAPI -> NASA FIRMS -> normalized hotspot JSON -> Leaflet markers

7 SEP MILESTONE

A real NOAA-20/NOAA-21 FIRMS thermal hotspot must appear on your own React map through your own FastAPI backend. If this fails, do not move to ML.

19.2 8 September - Phase 2: Geo-Context Engine

OMKAR

- Implement OSM / Overpass query around selected coordinates.
- Fetch industrial zones, works/factories, flares, power plants and quarries.
- Calculate nearest distances.
- Integrate land-cover lookup.
- Start PostGIS schema and persistence of key objects.

SHARI

- Build marker click interaction and hotspot detail drawer.
- Show raw satellite information.
- Show nearest industrial facility/context cards.
- Add map layer toggles.
- Build Analyze Location page.

8 SEP MILESTONE

Click a FIRMS hotspot -> AgniDrishti must answer: "What surrounds this thermal hotspot?" Show nearby industrial features and land-cover context.

19.3 9 September - Phase 3: AI Classifier

OMKAR

- Implement feature extraction.
- Create weak-label / silver-label dataset.
- Train Random Forest baseline.
- Generate real evaluation artifacts: classification report, confusion matrix, feature importance.
- Save model with Joblib.
- Implement classifier service returning class + probability.

SHARI

- Add classification badges / marker styling.
- Add confidence percentage.
- Build "Why this classification?" explanation UI.
- Wire analyzer response into hotspot drawer.

9 SEP MILESTONE

Click a hotspot -> the model returns a classification, probability and human-readable explanation. Do not fake evaluation metrics.

19.4 10 September - Phase 4: Differentiator

OMKAR

- Fetch historical detections around the same location. FIRMS area queries support historical dates in small day windows (commonly 1-5 days), so loop across windows for a 30-day history.
- Spatially cluster/associate repeat detections.
- Count unique detection dates.
- Compute historical median FRP.
- Compute persistence score.
- Compute FRP anomaly score.
- Integrate AgniRisk.

SHARI

- Build thermal history chart / panel.
- Show detections in 7 and 30 days.
- Show historical median FRP vs current FRP.
- Show Expected Persistent Source vs Abnormal Thermal Event status.
- Display AgniRisk prominently.

10 SEP MILESTONE

AgniDrishti must demonstrate the distinction between persistent industrial thermal activity and an abnormal industrial thermal event.

19.5 11 September - Phase 5: Freeze + Demo Preparation

Stop major development. Reliability now has higher value than another feature.

- Re-test every API and failure path.
- Choose and cache three strong demo locations/cases.
- Polish labels, spacing, loading states and error states.
- Finalize README and architecture/pipeline diagrams.
- Capture screenshots for PPT.
- Deploy and verify frontend/backend.
- Create offline/cached fallback.
- Record a backup screen capture of the complete demo.
- Rehearse the 4-minute story multiple times.

NO FEATURE CREEP

No "one more AI feature" on freeze day. New features can destroy a working prototype the night before the hackathon.

20. Three Demo Scenarios You Must Prepare

Case A - Persistent Industrial Source

- Choose a known refinery / flare / industrial context with repeated FIRMS detections.
- Expected result: Persistent Industrial Thermal Source.
- Show temporal history and relatively stable behavior.

Case B - Agricultural / Natural Fire

- Choose cropland or natural-land context with thermal activity and no nearby industrial infrastructure.
- Expected result: Agricultural Burning or Wildfire/Natural Fire.

Case C - Probable Abnormal Industrial Event

- Choose an industrial context with a transient / unusually strong thermal observation or low persistence.
- Expected result: Probable Industrial Thermal Event / Industrial Fire.

DEMO DISCIPLINE

Do not randomly click a hotspot during judging and hope it tells the right story. Prepare known coordinates whose context, history and classification you have already validated.

21. Offline / API Failure Fallback

Hackathon networks and public APIs can fail. Cache previously retrieved real observations and context so the demo remains reliable.

```
sample-data/
- demo_hotspots.json
- demo_osm_context.json
- demo_history.json
```

```
Environment flag
DEMO_MODE=true
```

- If live FIRMS/Overpass fails, fall back to cached real records.
- Do not fabricate fake "live" records.
- Judge-safe explanation: "For demonstration resilience, we cache the latest successfully retrieved observations; live mode consumes the APIs directly."

22. Do NOT Build Before 12 September

- Login / signup / user accounts
- Admin panel
- SMS alerting
- Android app
- Complicated authentication
- Full-country industrial master database
- CNN trained from scratch on Sentinel imagery
- LLM chatbot
- Multi-day fire forecasting
- Dozens of pages
- Blockchain
- Drone integration
- Weather prediction engine

These are future roadmap items at best. They consume sprint time without proving the core SIH26162 pipeline.

23. How AgniDrishti Should Differentiate

A generic "FIRMS + ML + map" story is not enough. Lead with the features that make the system operationally useful.

- 1. Temporal intelligence - persistent vs transient heat.
- 2. Abnormality detection - current FRP vs historical thermal baseline.
- 3. Explainable classification - show why the model predicted the class.
- 4. Probable source attribution - nearest industrial facility + distance.
- 5. Attribution uncertainty - explicit confidence; no unsupported exact causation.
- 6. Multi-source fusion - FIRMS + OSM + ESA WorldCover + historical FIRMS + imagery.

23.1 Competitive Positioning for SIH26162

Publicly visible SIH26162 team descriptions already mention combinations such as VIIRS thermal data, clustering/classification, industrial context, and live geospatial dashboards. Therefore, a basic "FIRMS + ML classifier + map" prototype is unlikely to stand out by itself.

- Lead with temporal persistence: expected industrial heat vs transient anomaly.
- Lead with FRP abnormality relative to historical baseline.
- Lead with explainability and uncertainty-aware probable source attribution.
- Demonstrate real integration and reliability rather than a slide-only architecture.

24. Judge-Criteria Mapping

Problem understanding	Thermal detection alone is not source classification; AgniDrishti adds context and segregation.
Novelty	Persistence + FRP anomaly + probable industrial-source attribution.
Technical depth	NASA FIRMS + OSM/Overpass + land cover + ML + PostGIS + GIS UI.
Feasibility	Software-first; largely open geospatial datasets and APIs; lightweight ML.
Prototype	Live/recent hotspot map with end-to-end analysis.
UX	One-click explainable hotspot analysis.
Scalability	Modular services, API separation and geospatial DB.
Future progression	Validated labels, analyst feedback loop, more sensors, alerts and agency integrations.

25. Four-Minute Internal Hackathon Demo Script

0:00-0:30 - Problem

Explain that satellite systems can detect thermal anomalies, but a hotspot does not by itself tell an analyst whether the source is wildfire, agricultural burning, gas flare or potentially abnormal industrial activity. Then say: "AgniDrishti gives that hotspot context and meaning."

0:30-1:15 - Live dashboard

- Open the dashboard.
- Show real/recent FIRMS detections.
- Toggle industrial / land-cover / imagery layers.
- Point out operational stats and filters.

1:15-2:10 - Persistent industrial source

- Click known persistent source.
- Show class + confidence.
- Show reasons: industrial context, repeated detections, stable FRP, flare/facility proximity.

2:10-3:00 - Abnormal industrial event

- Show low persistence / FRP spike / industrial context.
- Show probable industrial event and AgniRisk.
- Say: "Because satellite observations have finite spatial resolution, AgniDrishti provides probable nearby source attribution with confidence rather than claiming exact causation."

3:00-3:40 - Agricultural / wildfire case

- Show cropland/tree-cover context with no industrial features.

- Show corresponding classification and evidence.

3:40-4:00 - Close

CLOSING LINE

"NASA FIRMS tells us where the heat is. AgniDrishti tells us what that heat most likely means, why the model believes so, and whether industrial activity appears normal or anomalous."

26. PPT-to-Prototype Mapping

- Slide 1 - SIH26162 + AgniDrishti AI.
- Slide 2 - Proposed Solution: Detect -> Contextualize -> Classify -> Monitor.
- Slide 3 - Technical Approach: FIRMS -> OSM + WorldCover -> features -> AI -> persistence -> PostGIS -> dashboard. Include prototype screenshot.
- Slide 4 - Feasibility: open datasets, open geospatial stack, near-real-time API, no specialized hardware, modular architecture.
- Slide 5 - Impact: faster anomaly triage, industrial risk monitoring, reduced false attribution, persistent source tracking, analyst decision support.
- Slide 6 - References: NASA FIRMS, ESA WorldCover, OpenStreetMap/Overpass, VIIRS documentation.

27. Final 11 September Freeze Checklist

- ☐ Live NASA FIRMS hotspots
- ☐ NOAA-20 / NOAA-21 integration
- ☐ OSM industrial context
- ☐ ESA land-cover context
- ☐ Satellite/base imagery
- ☐ GIS map
- ☐ PostgreSQL/PostGIS
- ☐ Hotspot classification
- ☐ AI probability
- ☐ Nearby facility attribution
- ☐ Explanation panel
- ☐ Temporal persistence
- ☐ FRP anomaly
- ☐ AgniRisk
- ☐ Three rehearsed demo coordinates
- ☐ Offline cached fallback
- ☐ README
- ☐ Architecture diagram
- ☐ Clean Git commit history
- ☐ Deployed application
- ☐ SIH PPT
- ☐ Backup screen recording

28. Team Ownership

Owner	Primary responsibility	Deliverables
SHARI	Product + GIS frontend + integration	React app, dashboard, Leaflet, markers, filters, analysis drawer, charts, coordinate search, API integration, demo polish, PPT screenshots
OMKAR	Data + backend + AI	FastAPI, FIRMS, OSM/Overpass, WorldCover, feature engineering, persistence, Random Forest, AgniRisk, PostGIS, endpoints

INTEGRATE EVERY NIGHT

Do not build frontend and backend independently until Friday. Integrate Sep 7, Sep 8, Sep 9, Sep 10, and freeze on Sep 11.

29. What the Final AgniDrishti Pitch Should Sound Like

SIH-LEVEL PRODUCT DESCRIPTION

AgniDrishti is an explainable geospatial intelligence platform that fuses NASA VIIRS thermal detections, OpenStreetMap industrial infrastructure, Sentinel-derived ESA land cover, and temporal hotspot behavior to classify thermal anomalies and distinguish persistent industrial heat from potentially abnormal industrial events.

29.1 Simple mental model

HOTSPOT

- > What is near it?
- > What type of land is it on?
- > Has it happened before?
- > Is today's heat abnormal?
- > What does the model think?
- > Why?
- > Should an analyst investigate?

30. First Non-Negotiable Milestone

BY END OF 7 SEPTEMBER

A real NOAA-20/NOAA-21 FIRMS hotspot must appear on your own React map through your own FastAPI backend. Once that works, layer geospatial context, ML, persistence and risk on top of it.

31. Reference Links / Technical Reading

NASA FIRMS API: <https://firms.modaps.eosdis.nasa.gov/api/>

NASA FIRMS Area API: <https://firms.modaps.eosdis.nasa.gov/api/area/>

NASA FIRMS API Academy / field guidance:

https://firms.modaps.eosdis.nasa.gov/content/academy/data_api/firms_api_use.html

NASA FIRMS Map / detection context: <https://firms.modaps.eosdis.nasa.gov/map/>

OpenStreetMap man_made=works: https://wiki.openstreetmap.org/wiki/Tag:man_made=works

ESA WorldCover data access: <https://esa-worldcover.org/en/data-access>

ESA WorldCover v200 catalog:

https://developers.google.com/earth-engine/datasets/catalog/ESA_WorldCover_v200

Terrascope OGC services: <https://docs.terrascope.be/Developers/WebServices/OGC/WMTSv2.html>

NMIT Institution Innovation Council / internal hackathon context: <https://nitte.edu.in/nmit/iic.php>

Public SIH26162 team activity example: <https://in.linkedin.com/in/jupallevenkat>

Example SIH internal hackathon template resource: <https://sih-internalhackathon-ilead.pages.dev/>

32. Final Build Principle

WORKING END-TO-END PIPELINE > EXTRA FEATURES

If the five core layers work reliably - real hotspot ingestion, geospatial context, explainable classification, temporal persistence, and judge-friendly visualization - AgniDrishti will look like a serious SIH prototype rather than a rushed student dashboard.